THE PRIMITIVE AND DIAMOND SURFACES LOCALLY MINIMIZE THE VARIANCE OF GAUSS CURVATURE

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ABSTRACT. We prove that Schwarz' Primitive and Diamond surfaces are local minimizers for the variance of Gauss curvature among local deformations within the space of Triply Periodic Minimal Surfaces of genus 3. This is done by viewing the branched values of the Gauss map as a point configuration on the sphere, and rewriting the variance of Gauss curvature as a product of two integrals involving exponentials of Green functions.

Triply Periodic Minimal Surfaces (TPMSs) are minimal surfaces in flat 3-tori. The genus of a TPMS is at least three. A TPMS of genus three (TPMSg3) has a Weierstrass parameterization defined on a branched double cover of the sphere with eight branch points. The branched covering map is provided by the Gauss map, so the branch points correspond to eight *flat points* on the TPMS (where the Gauss curvature vanishes). We refer the readers to [?] for general reference.

TPMSs find applications in natural sciences as a model for periodic bicontinuous structures [?]. In nature and in laboratories, the chance to observe Schwarz' Primitive (P), Diamond (D) and Schoen's Gyroid (G) surfaces is much higher than finding other TPMSs. Note that these surfaces belong to the same Bonnet family. They are therefore locally isometric, have the same branch points, and the same variance of Gauss curvature. For all the three surfaces, the branch points are vertices of a cube. Based on numerical evidence [?], it was conjectured that

Conjecture. *Schwarz' Primitive, Diamond, and Schoen's Gyroid surfaces have the minimal variance of Gauss curvature among all Triply Periodic Minimal Surfaces of genus 3.*

In this short note, we prove that the Primitive and Diamond are indeed local minimizers.

Theorem. *Schwarz' Primitive and Diamond surfaces are local minimizers for the variance of Gauss curvature among local deformations within the space of Triply Periodic Minimal Surfaces of genus 3.*

We will relax the problem to study the variance for the Gauss curvature defined on the 13-dimensional space V of configurations of eight points on the sphere up to global rotations. They correspond to the normal vectors at the eight flat points on a TPMSg3. For Diamond, Primitive, and Gyroid surfaces, the eight points are at the vertices of a cube. However, only a 5-dimensional variety in V correspond to TPMSg3s. In particular, the 5-dimensional subspace of antipodal configurations correspond to embedded TPMSg3s known as the *Meeks family*.

In [?], it was proved that, up to homotheties, the Primitive, Diamond, and Gyroid all belong to a unique locally rigid 5-parameter smooth family of pairwise non-homothetic triply periodic minimal surfaces. In particular, the family that

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Date: April 28, 2026.

2010 *Mathematics Subject Classification.* Primary 53A10.

Key words and phrases. minimal surfaces.

contains the Diamond and the Primitive surfaces is the Meeks family. Schoen's Gyroid, however, does not belong to Meeks family.

At the cubic configuration, we compute the Hessian of the variance for the Gauss curvature on V and find, unfortunately, a 2-dimensional negative eigenspace generated by "twists". So the Hessian is indefinite, and the cubic configuration is not a local minimizer in V. Nevertheless, we are able to prove positive definiteness for the deformations preserving antipodality, so the cubic configuration is a local minimizer among antipodal deformations, thereby proving the Theorem.

The 5-parameter family of local deformations of the Gyroid is not fully understood. Among the known deformations of the Gyroid, it is indeed a local minimizer. But in the presence of the negative eigenspace, we can not rule out a possibility that it might not be a local minimizer along other deformations.

Acknowledgement. The author thanks Chengjian Yao and Thierry De Pauw for very helpful and inspiring discussions.

1. Reformulation

1.1. **Weierstrass parameterization.** Given a TPMSg3, let $p_1, \cdots, p_8$ be (the stereographic projection of) the Gauss map at the eight flat points. Consider the hyperelliptic Riemann surface Σ of genus three defined by

$$w^2 = P(z) = \prod_{i=1}^{8}(z - p_i).$$

It can be seen as a branched double cover of the Riemann sphere $\mathbb{S}^2 \simeq \mathbf{CP}^1$ with eight branch points. Then we have the following *Weierstrass parameterization* for the TPMS (up to translations)

$$\Sigma \ni (z, w) \mapsto \operatorname{Re} \int^{(z,w)} \frac{(1 - z^2, \mathrm{i}(1 + z^2), 2z)}{w} e^{i\vartheta}\, dz. \tag{1}$$

Switching the two branches by $(z, w) \mapsto (z, -w)$ correspond to the inversion with respect to any of the eight flat points. The parameter ϑ is the *Bonnet angle*; changing ϑ gives a locally isometric deformation of the surface known as *Bonnet rotation*. Two surfaces whose ϑ differ by $\pi/2$ are called *conjugate pairs*.

For the minimal surface to be triply periodic and globally well-defined, one needs to solve the period problem (aka, *close the periods*). More precisely, one finds p_i's and ϑ so that the real parts of the integrals over closed cycles on Σ form a three dimensional lattice. The period problem involves three real equations for each of the three coordinates, hence nine real equations in total. Since we have 17 real parameters (p_i's and ϑ), the set of TPMSg3s form a 8-dimensional variety up to translations, or 5-dimensional variety up to homotheties.

Closing the periods is usually very difficult. But Meeks proved in [**?**] that, when the branch points form four antipodal pairs, there exists a conjugate pair embedded TPMSg3s. Up to homotheties, these TPMSg3s form a 5-dimensional manifold termed *Meeks family*.

In this paper, we will not insist on closing the periods. The variance of Gauss curvature is invariant under local isometries such as Bonnet rotations. So it is a function defined on a 13-dimensional space of configurations of eight points on the sphere up to rotations. The Meeks surfaces correspond to the 5-dimensional subspace of antipodal configurations, for which we know the existence of ϑ that closes the periods. Our main Theorem is equivalent to that the cubic configuration is a local minimizer of the Gauss variance on this 5-dimensional subspace.

1.2. **Variance of Gauss curvature.** More precisely, we want to minimize the (dimensionless) variance of Gauss curvature, defined by

$$\sigma(K) = \frac{\langle K^2\rangle - \langle K\rangle^2}{\langle K\rangle^2} = \left[\frac{\int_\Sigma K^2\,dA}{\int_\Sigma dA} - \left(\frac{\int_\Sigma K\,dA}{\int_\Sigma dA}\right)^2\right] \Big/ \left(\frac{\int_\Sigma K\,dA}{\int_\Sigma dA}\right)^2.$$

By Gauss-Bonnet, $\int_\Sigma K\,dA$ is a topological invariant. It is then equivalent to minimize

$$\int_\Sigma K^2 dA \times \int_\Sigma dA. \tag{2}$$

From the Weierstrass parameterization, we compute

$$dA = \frac{(1+|z|^2)^2}{|w|^2}\,dx\wedge dy, \qquad K = -4\frac{|w|^2}{(1+|z|^2)^4},$$

where $z = x + y\mathrm{i}$. Indeed, one verifies that

$$\int_\Sigma K dA = -4\int_\Sigma \frac{dx\wedge dy}{(1+|z|^2)^2} = -8\pi$$

as expected. The area element and the Gauss curvature are invariant after switching the branches. It then suffices to compute the invariance on one of the branch. So (2) becomes (up to a constant factor)

$$\int_{\mathbf{CP}^1} \frac{|P(z)|}{(1+|z|^2)^6}\,dx\wedge dy \times \int_{\mathbf{CP}^1} \frac{(1+|z|^2)^2}{|P(z)|}\,dx\wedge dy. \tag{3}$$

Note that the second integral is improper: The integrand is not defined at the zeros of $P(z)$. So the integral is defined as the limit

$$\int_{\mathbf{CP}^1} \frac{(1+|z|^2)^2}{|P(z)|}\,dx\wedge dy = \lim_{\varepsilon\to 0}\int_{\mathbf{CP}^1\setminus\cup_i D^i_\varepsilon} \frac{(1+|z|^2)^2}{|P(z)|}\,dx\wedge dy,$$

where D^i_ε denote the ε-disks around p_i under the Fubini–Study metric. Fortunately, the limit converges.

1.3. **Fubini–Study metric.** Note that

$$\mathrm{d}A_{\mathrm{FS}} = \frac{4\,dx\wedge dy}{(1+|z|^2)^2}$$

is the area element for the Fubini–Study metric on $\mathbf{CP}^1$, which is (normalized to) exactly the standard metric on the unit sphere $\mathbb{S}^2$. Moreover, we can identify $P(z)$ to a global section s of the holomorphic line bundle $\mathcal{O}(8) = \mathcal{O}(1)^{\otimes 8}$ over $\mathbf{CP}^1$, and

$$\|s\|_{\mathrm{FS}} = \frac{|P(z)|}{(1+|z|^2)^4}$$

is its point-wise Fubini–Study norm. See, for instance, [**?**]. Then (3) becomes (up to a constant factor)

$$J_8 := \int_{\mathbf{CP}^1} \|s\|_{\mathrm{FS}}\,\mathrm{d}A_{\mathrm{FS}} \times \int_{\mathbf{CP}^1} \|s\|^{-1}_{\mathrm{FS}}\,\mathrm{d}A_{\mathrm{FS}}. \tag{4}$$

Note again that the second integral is improper. It is defined as the convergent limit of the integral over $\mathbf{CP}^1 \setminus \cup_i D^i_\varepsilon$ as $\varepsilon \to 0$.

1.4. **Lelong–Poincaré Equation.** Define

$$u = \log \|s\|_{\mathrm{FS}}.$$

Note that the Fubini–Study metric on $\mathcal{O}(8)$ is Hermitian. One then easily verifies the Lelong–Poincaré Equation

$$\Delta u = 2\pi \sum_{i=1}^{8} \delta(z - p_i) - 4\frac{4}{(1+|z|^2)^2},$$

or, using the Laplacian under the Fubini–Study metric,

$$\Delta_{\mathrm{FS}} u = \frac{(1+|z|^2)^2}{4} \Delta u = 2\pi \sum_{i=1}^{8} \delta_i(z) - 4,$$

where

$$\delta_i(z) = \frac{(1+|z|^2)^2}{4} \delta(z - p_i)$$

denote the Dirac delta function under the Fubini–Study metric centered at p_i.

The solution is given by

$$u = 2\pi \sum_{i=1}^{8} G_i(z),$$

where $G_i(z)$ is the Green function solving

$$\Delta_{\mathrm{FS}} G_i(z) = \delta_i(z) - \frac{1}{4\pi}. \tag{5}$$

On the unit sphere $\mathbb{S}^2 \simeq \mathbf{CP}^1$, we have

$$G_i(z) = -\frac{1}{2\pi} \log \sin \frac{\mathrm{dist}(z, p_i)}{2} + C$$

where $\mathrm{dist}(z, p_i)$ is the spherical distance between z and p_i. Then (4) becomes

$$J_8 = \int_{\mathbb{S}^2} \exp(u)\, \mathrm{d}A_{\mathbb{S}^2} \times \int_{\mathbb{S}^2} \exp(-u)\, \mathrm{d}A_{\mathbb{S}^2}, \tag{6}$$

where $\mathrm{d}A_{\mathbb{S}^2}$ is the standard spherical area element on the unit sphere. Note that (6) does not depend on the constant C in G_i, so we assume $C = 0$ from now on. Note again that the second integral is improper and is defined as the convergent limit

$$\int_{\mathbb{S}^2} \exp(-u)\, \mathrm{d}A_{\mathbb{S}^2} = \lim_{\varepsilon \to 0} \int_{\Omega_\varepsilon} \exp(-u)\, \mathrm{d}A_{\mathbb{S}^2},$$

where

$$\Omega_\varepsilon = \mathbb{S}^2 \setminus \cup_{i=1}^{n} D_\varepsilon^i.$$

The Conjecture would be proved if J_8 is minimized when $p_i \in \mathbb{S}^2$ are the vertices of a cube.

Remark 1. With this reformulation, we actually expand our consideration beyond TPMSg3s. We are now considering all minimal surfaces with the Weierstrass parameterization (1). In particular, we do not need the period problems to be solved, not to mention the embeddedness.

1.5. **Generalization.** The analysis above can be generalized as follows. Fix n points $p_1, \cdots, p_n$, and define the polynomial

$$P(z) = \prod_{i=1}^{n} (z - p_i).$$

Consider the product of integrals

$$\int_{\mathbf{CP}^1} \frac{|P(z)|}{(1+|z|^2)^{\frac{n}{2}+2}}\, dx \wedge dy \times \int_{\mathbf{CP}^1} \frac{(1+|z|^2)^{\frac{n}{2}-2}}{|P(z)|}\, dx \wedge dy.$$

Note that

$$\|s\|_{\mathrm{FS}} = \frac{|P(z)|}{(1+|z|^2)^{n/2}}$$

is the point-wise Fubini–Study norm of $P(z)$ as a global section s of $\mathcal{O}(n)$ over $\mathbf{CP}^1$. Then the product can be rewritten (up to a constant factor) into the form

$$J_n := \int_{\mathbf{CP}^1} \|s\|_{\mathrm{FS}}\, \mathrm{d}A_{\mathrm{FS}} \times \int_{\mathbf{CP}^1} \|s\|_{\mathrm{FS}}^{-1}\, \mathrm{d}A_{\mathrm{FS}}.$$

Now the Lelong–Poincaré Equation is

$$\Delta_{\mathrm{FS}} u = 2\pi \sum_{i=1}^{n} \delta_i(z) - n/2,$$

which is solved by

$$u = 2\pi \sum_{i=1}^{n} G_i(z).$$

So

$$J_n = \int_{\mathbb{S}^2} \exp(u)\, \mathrm{d}A_{\mathbb{S}^2} \times \int_{\mathbb{S}^2} \exp(-u)\, \mathrm{d}A_{\mathbb{S}^2}.$$

And we could ask the following problem

Problem. *What are the positions of $p_1, \cdots, p_n$ that minimize J_n?*

When $n = 8$, this is exactly the problem of finding branch values to minimize the variance of the Gauss curvature.

2. Gradient and Hessian

2.1. **The gradient.** Recall that

$$\Omega_\varepsilon = \mathbb{S}^2 \setminus \cup_{i=1}^{n} D_\varepsilon^i,$$

where D_ε^i denote the ε-disks around p_i under the spherical metric. Define

$$I_\varepsilon^\pm = \int_{\Omega_\varepsilon} \exp(\pm u)\, \mathrm{d}A_{\mathbb{S}^2},$$

and two probability measure $\mathrm{d}\mu_\varepsilon^\pm = \rho_\varepsilon^\pm \mathrm{d}A_{\mathbb{S}^2}$ on Ω_ε, where

$$\rho_\varepsilon^\pm = \frac{\exp(\pm u)}{\int_{\Omega_\varepsilon} \exp(\pm u)\, \mathrm{d}A_{\mathbb{S}^2}} = \frac{\exp(\pm u)}{I_\varepsilon^\pm}.$$

We compute

$$\begin{aligned}
\langle \nabla_{p_i} \log I_\varepsilon^\pm, v_i \rangle &= \frac{\langle \nabla_{p_i} I_\varepsilon^\pm, v_i \rangle}{I_\varepsilon^\pm} \\
&= \frac{1}{I_\varepsilon^\pm} \Big(\int_{\Omega_\varepsilon} \langle \pm 2\pi \nabla_{p_i} G_i, v_i \rangle \exp(\pm u)\, \mathrm{d}A_{\mathbb{S}^2} + \oint_{\partial D_\varepsilon^i} \exp(\pm u) \langle n_i, \tilde{v}_i \rangle \mathrm{d}s \Big) \\
&= \int_{\Omega_\varepsilon} \langle \pm 2\pi \nabla_{p_i} G_i, v_i \rangle \rho_\varepsilon^\pm\, \mathrm{d}A_{\mathbb{S}^2} + \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i \rangle \rho_\varepsilon^\pm \mathrm{d}s
\end{aligned}$$

where $v_i \in T_{p_i}\mathbb{S}^2$, n_i is the unit normal vectors on the boundary of D^i_ε pointing to the interiors of the disks, and $\tilde{v}_i$ is a Killing field such that $\tilde{v}_i(p_i) = v_i$. So for

$$v = (v_1, \cdots, v_8) \in \bigoplus_{i=1}^{8} T_{p_i}\mathbb{S}^2,$$

we have

$$\langle \nabla_p \log I^\pm_\varepsilon, v\rangle = \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi\nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu^\pm_\varepsilon + \sum_{i=1}^n \oint_{\partial D^i_\varepsilon} \langle n_i, \tilde{v}_i\rangle \rho^\pm_\varepsilon \mathrm{d}s \tag{7}$$

Note that the formula does not depend on the choice of the Killing fields $\tilde{v}_i$. To see this, assume another choice $\tilde{v}'_i$. Then $\tilde{w}_i = \tilde{v}'_i - \tilde{v}_i$ is a Killing field that vanishes at p_i, which is given as a rotation around p_i. Consequently, we have $\langle \mp 2\pi\nabla_z G_i, \tilde{w}_i\rangle = 0$ and $\langle n_i, \tilde{w}_i\rangle = 0$.

Note also that, when $\tilde{v}_i = \tilde{v}_*$ is the same Killing field for all i, then

$$\begin{aligned}\langle \nabla_p \log I^\pm_\varepsilon, v\rangle &= \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi\nabla_z G_i, \tilde{v}_*\rangle \mathrm{d}\mu^\pm_\varepsilon + \sum_{i=1}^n \oint_{\partial D^i_\varepsilon} \langle n_i, \tilde{v}_*\rangle \rho^\pm_\varepsilon \mathrm{d}s\\ &= \int_{\Omega_\varepsilon} \langle \mp\nabla_z u, \tilde{v}_*\rangle \mathrm{d}\mu^\pm_\varepsilon + \int_{\Omega_\varepsilon} \langle \pm\nabla_z u, \tilde{v}_*\rangle \mathrm{d}\mu^\pm_\varepsilon = 0\end{aligned}$$

So a deformation along a Killing field does not change $I^\pm_\varepsilon$.

Recall that $2\pi G_i = \log\sin(\frac{1}{2}\operatorname{dist}(z, p_i))$, so

$$2\pi\nabla_z G_i \sim \frac{\hat{w}_i}{\operatorname{dist}(z, p_i)}, \qquad \rho^\pm_\varepsilon \sim \operatorname{dist}(z, p_i)^{\pm 1}$$

as $z \to p_i$, where $\hat{w}_i = \nabla_z \operatorname{dist}(z, p_i)$. It is then obvious that (7) converges as $\varepsilon \to 0$ for μ^+_ε. As for μ^-_ε, in the standard spherical coordinate with the pole p_i, polar angle θ and azimuth angle ϕ, the integrands

$$\begin{aligned}\langle 2\pi\nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu^-_\varepsilon &\sim \frac{2}{\theta I^-}\langle \hat{w}_i(\theta,\phi), \tilde{v}_i(\theta,\phi)\rangle \mathrm{d}\theta\mathrm{d}\phi,\\ \langle n_i, \tilde{v}_i\rangle \rho^-_\varepsilon \mathrm{d}s &\sim \frac{-2}{I^-}\langle \hat{w}_i(\varepsilon,\phi), \tilde{v}_i(\varepsilon,\phi)\rangle \mathrm{d}\phi\end{aligned}$$

as $\theta, \varepsilon \to 0$. The integrals in (7) then converge as $\varepsilon \to 0$ because

$$\lim_{\theta\to 0} \int_0^{2\pi} \langle \hat{w}_i(\theta,\phi), \tilde{v}_i(\theta,\phi)\rangle \mathrm{d}\phi \to 0.$$

More specifically, we have

$$\langle \nabla_p \log I^\pm, v\rangle := \lim_{\varepsilon\to 0}\langle \nabla_p \log I^\pm_\varepsilon, v\rangle = \int_{\mathbb{S}^2} \sum_{i=1}^n \langle \mp 2\pi\nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu^\pm.$$

As a sanity check, one easily verifies that this vanishes when $\tilde{v}_i = \tilde{v}_*$ is the same Killing field for all i.

Finally, by symmetry, one easily verifies that $\langle \nabla_p \log I^\pm_\varepsilon, v\rangle = 0$ for any v when $p_1, \cdots, p_8$ are the vertices of the cube. So the cube is indeed a critical point of J_8.

2.2. **The Hessian.** We compute,

$$\begin{aligned}\langle \nabla_{p_i}\rho^\pm_\varepsilon, v_i\rangle &= \frac{\langle \pm 2\pi\nabla_{p_i} G_i, v_i\rangle \exp(\pm u)}{I^\pm_\varepsilon} - \frac{\exp(\pm u)\langle \nabla_{p_i} I^\pm_\varepsilon, v_i\rangle}{(I^\pm_\varepsilon)^2}\\ &= \rho^\pm_\varepsilon\Big(\langle \mp 2\pi\nabla_z G_i, \tilde{v}_i\rangle - \int_{\Omega_\varepsilon} \langle \mp 2\pi\nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu^\pm_\varepsilon - \oint_{\partial D^i_\varepsilon} \langle n_i, \tilde{v}_i\rangle \rho^\pm_\varepsilon \mathrm{d}s\Big)\end{aligned}$$

The Hessian

$$
\begin{aligned}
\nabla_p^2 \log I_\varepsilon^\pm(v,v) =& \langle \nabla_p \langle \nabla_p \log I_\varepsilon^\pm, v\rangle, v\rangle \\
=& \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \nabla_{p_i} \langle \pm 2\pi \nabla_{p_i} G_i, v_i\rangle, v_i\rangle \rho_\varepsilon^\pm \, \mathrm{d}A_{\mathbb{S}^2} \qquad (8)\\
&+ \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \pm 2\pi \nabla_{p_i} G_i, v_i\rangle \sum_{j=1}^n \langle \nabla_{p_j} \rho_\varepsilon^\pm, v_j\rangle \, \mathrm{d}A_{\mathbb{S}^2} \qquad (9)\\
&+ \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i\rangle \sum_{j=1}^n \langle \nabla_{p_j} \rho_\varepsilon^\pm, v_j\rangle \mathrm{d}s \qquad (10)\\
&+ \sum_{j=1}^n \oint_{\partial D_\varepsilon^j} \langle n_j, \tilde{v}_j\rangle \sum_{i=1}^n \langle \pm 2\pi \nabla_{p_i} G_i, v_i\rangle \rho_\varepsilon^\pm \, \mathrm{d}s \qquad (11)\\
&+ \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i\rangle \langle \nabla_z \rho_\varepsilon^\pm, \tilde{v}_i\rangle \mathrm{d}s
\end{aligned}
$$

We expand each term

$$
(8) = \int_{\Omega_\varepsilon} \sum_{i=1}^n \pm 2\pi \nabla_z^2 G_i(\tilde{v}_i, \tilde{v}_i) \mathrm{d}\mu_\varepsilon^\pm
$$

$$
\begin{aligned}
(9) =& \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi \nabla_z G_i, \tilde{v}_i\rangle \sum_{j=1}^n \langle \nabla_{p_j} \rho_\varepsilon^\pm, v_j\rangle \, \mathrm{d}A_{\mathbb{S}^2} \\
=& \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi \nabla_z G_i, \tilde{v}_i\rangle \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde{v}_j\rangle \mathrm{d}\mu_\varepsilon^\pm \\
&- \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi \nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu_\varepsilon^\pm \int_{\Omega_\varepsilon} \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde{v}_j\rangle \mathrm{d}\mu_\varepsilon^\pm \\
&- \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi \nabla_z G_i, \tilde{v}_i\rangle \mathrm{d}\mu_\varepsilon^\pm \sum_{j=1}^n \oint_{\partial D_\varepsilon^j} \langle n_j, \tilde{v}_j\rangle \rho_\varepsilon^\pm \, \mathrm{d}s
\end{aligned}
$$

$$
\begin{aligned}
(10) =& \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i\rangle \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde{v}_j\rangle \rho_\varepsilon^\pm \, \mathrm{d}s \\
&- \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i\rangle \rho_\varepsilon^\pm \, \mathrm{d}s \int_{\Omega_\varepsilon} \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde{v}_j\rangle \mathrm{d}\mu_\varepsilon^\pm \\
&- \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde{v}_i\rangle \rho_\varepsilon^\pm \, \mathrm{d}s \sum_{j=1}^n \oint_{\partial D_\varepsilon^j} \langle n_j, \tilde{v}_j\rangle \rho_\varepsilon^\pm \, \mathrm{d}s,
\end{aligned}
$$

and

$$
(11) = \sum_{j=1}^n \int_{\partial D_\varepsilon^j} \langle n_j, \tilde{v}_j\rangle \sum_{i=1}^n \langle \mp 2\pi \nabla_z G_i, \tilde{v}_i\rangle \rho_\varepsilon^\pm \, \mathrm{d}s.
$$

Putting them together gives

$$\nabla_p^2 \log I_\varepsilon^\pm(v,v) = \int_{\Omega_\varepsilon} \Big(\sum_{i=1}^n \langle 2\pi\nabla_z G_i, \tilde v_i\rangle\Big)^2 \mathrm{d}\mu_\varepsilon^\pm \tag{12}$$

$$+ \int_{\Omega_\varepsilon} \sum_{i=1}^n \pm 2\pi \nabla_z^2 G_i(\tilde v_i, \tilde v_i) \mathrm{d}\mu_\varepsilon^\pm \tag{13}$$

$$- \Big(\int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \mp 2\pi\nabla_z G_i, \tilde v_i\rangle \mathrm{d}\mu_\varepsilon^\pm + \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde v_i\rangle \rho_\varepsilon^\pm \mathrm{d}s\Big)^2 \tag{14}$$

$$+ \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde v_i\rangle \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde v_j\rangle \rho_\varepsilon^\pm \,\mathrm{d}s \tag{15}$$

$$+ \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde v_i\rangle \sum_{j=1}^n \langle \mp 2\pi \nabla_z G_j, \tilde v_j - \tilde v_i\rangle \rho_\varepsilon^\pm \,\mathrm{d}s \tag{16}$$

Note again the independence to the choice of the Killing fields $\tilde v_i$ (by the same argument as before) and, when $\tilde v_i = \tilde v_*$ is the same Killing field for all i,

$$\begin{aligned}\nabla_p^2 \log I_\varepsilon^\pm(v,v) = \int_{\Omega_\varepsilon} \langle \nabla_z u, \tilde v_*\rangle^2 \mathrm{d}\mu_\varepsilon^\pm + \int_{\Omega_\varepsilon} \pm \nabla_z^2 u(\tilde v_*, \tilde v_*) \mathrm{d}\mu_\varepsilon^\pm \\ + \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde v_*\rangle \langle \mp \nabla_z u, \tilde v_*\rangle \rho_\varepsilon^\pm \,\mathrm{d}s = 0.\end{aligned}$$

The boundary term (16) converges to 0 because D_ε^i only contains the singularity of G_j when $j = i$. For the same reason, in the limit $\varepsilon \to 0$, we may rewrite the "coupling" boundary term (15) into the form

$$\begin{aligned}&\lim_{\varepsilon\to 0} \sum_{i=1}^n \oint_{\partial D_\varepsilon^i} \langle n_i, \tilde v_i\rangle \langle \mp 2\pi \nabla_z G_i, \tilde v_i\rangle \rho_\varepsilon^\pm \,\mathrm{d}s \\ =&\lim_{\varepsilon\to 0} \sum_{i=1}^n \Big[-\int_{\Omega_\varepsilon} \langle \nabla_z u, \tilde v_i\rangle \langle 2\pi \nabla_z G_i, \tilde v_i\rangle \,\mathrm{d}\mu_\varepsilon^\pm \mp \int_{\Omega_\varepsilon} 2\pi \nabla_z^2 G_i(\tilde v_i, \tilde v_i)\,\mathrm{d}\mu_\varepsilon^\pm\Big],\end{aligned}$$

where the second term cancels with (13).

We have seen that the second term in (14) converges to 0 as $\varepsilon \to 0$. At a critical point, its first term (14) vanishes. If this is the case, we have

$$\begin{aligned}&\lim_{\varepsilon\to 0} \nabla_p^2 \log I_\varepsilon^\pm(v,v) \\ =&\lim_{\varepsilon\to 0} \int_{\Omega_\varepsilon} \Big(\sum_{i=1}^n \langle 2\pi \nabla_z G_i, \tilde v_i\rangle\Big)^2 \mathrm{d}\mu_\varepsilon^\pm - \int_{\Omega_\varepsilon} \sum_{i=1}^n \langle \nabla_z u, \tilde v_i\rangle \langle 2\pi \nabla_z G_i, \tilde v_i\rangle \,\mathrm{d}\mu_\varepsilon^\pm \\ =&\lim_{\varepsilon\to 0} \int_{\Omega_\varepsilon} \sum_{i=1}^n \sum_{j=1}^n \langle 2\pi \nabla_z G_j, \tilde v_j - \tilde v_i\rangle \langle 2\pi \nabla_z G_i, \tilde v_i\rangle \,\mathrm{d}\mu_\varepsilon^\pm.\end{aligned}$$

When $i \neq j$, the convergence follows from the same argument for the convergence of the gradience (7). The possible divergence with $i = j$ is ruled out by $\tilde v_j - \tilde v_i$. In the following, we will simply write

$$\nabla_p^2 \log I^\pm(v,v) = \int_{\mathbb{S}^2} \sum_{i=1}^n \sum_{j=1}^n \langle 2\pi \nabla_z G_j, \tilde v_j - \tilde v_i\rangle \langle 2\pi \nabla_z G_i, \tilde v_i\rangle \,\mathrm{d}\mu^\pm.$$

3. Local minimizer

Without loss of generality, we assume that the vertices of the cube, in the standard coordinates (θ, ϕ) of the sphere, are as follows:

$$
\begin{aligned}
p_1 &= (\theta_0, 0), & p_5 &= (\pi - \theta_0, 0),\\
p_2 &= (\theta_0, \pi/2), & p_6 &= (\pi - \theta_0, \pi/2),\\
p_3 &= (\theta_0, \pi), & p_7 &= (\pi - \theta_0, \pi),\\
p_4 &= (\theta_0, 3\pi/2), & p_8 &= (\pi - \theta_0, 3\pi/2).
\end{aligned}
$$

where $\theta_0 = \arccos(1/\sqrt{3})$.

3.1. **Decomposition into irreducible representations.** We have seen that the Hessian vanishes on Killing fields, compatible with the fact the J is invariant under global rotations. So we study the Hessian on the 13-dimensional space

$$V = \Big(\bigoplus_{i=1}^{8} T_{p_i}\mathbb{S}^2\Big)/\mathrm{SO}(3)$$

at the cubic configuration. For this purpose, we decompose V into irreducible representations (irreps) of the symmetry group of the cube, as follows:

- The 3-dimensional space T_{1u} generated by slides (that moves the center of mass). One generator is given by

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1, 2, 5, 6\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{3, 4, 7, 8\}.$$

- The 2-dimensional space E_g generated by orbit of

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1, 3, 5, 7\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{2, 4, 6, 8\}.$$

  Every face remain rectangular under these deformations.
- The 2-dimensional space E_u generated by orbit of

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1, 2, 3, 4\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{5, 6, 7, 8\}.$$

  The face centers of the cube remain symmetry centers under these deformations.
- The 3-dimensional space T_{2g} generated by the orbit of

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1, 2, 7, 8\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{3, 4, 5, 6\}.$$

- The 3-dimensional space T_{2u} generated by orbit of

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1, 3, 6, 8\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{2, 4, 5, 7\}.$$

The Hessian is invariant under the action of G. By Schur's lemma, it must be a multiple of identity on each of the irreps. So we only need to verify that the Hessian is non-negative on one generator for each of the irreps.

3.2. **The definiteness on T_{1u}, E_u and E_g.** We have chosen the generator vectors, intentionally, with the following property. The eight vertices are partitioned into two subsets A and B, with four vertices each, such that vertices in A extend to the Killing field ∂_ϕ, while vertices in B extend to $-\partial_\phi$. In this case, we have

$$\nabla_p^2 \log I^\pm(v,v) = \int_{\mathbb{S}^2} \sum_{i=1}^n \sum_{j=1}^n \langle 2\pi \nabla_z G_j, \tilde{v}_j - \tilde{v}_i \rangle \langle 2\pi \nabla_z G_i, \tilde{v}_i \rangle \, \mathrm{d}\mu^\pm$$
$$= -4 \int_{\mathbb{S}^2} \sum_{i\in A} \sum_{j \in B} (2\pi \partial_\phi G_j)(2\pi \partial_\phi G_i) \, \mathrm{d}\mu^\pm = -16\pi^2 \int_{\mathbb{S}^2} \partial_\phi G_A \partial_\phi G_B \, \mathrm{d}\mu^\pm,$$

where

$$G_A = \sum_{i\in A} G_i, \qquad G_B = \sum_{i \in B} G_i.$$

To study the definiteness of the Hessian, we need to first study the sign of

$$H_{ij}^\pm = \int_{\mathbb{S}^2} \partial_\phi G_i \partial_\phi G_j \, \mathrm{d}\mu^\pm.$$

When p_i and p_j are antipodal (e.g. $(i,j) = (1,7)$) or on a horizontal face diagonal (e.g. $(i,j) = (1,3)$), one easily sees that the integrand $\partial_\phi G_i \partial_\phi G_j \le 0$ everywhere on the sphere, so $H_{ij}^\pm$ is negative. When p_i and p_j are vertical neighbors (e.g. $(i,j) = (1,5)$), one easily sees that the integrand $\partial_\phi G_i \partial_\phi G_j \ge 0$ everywhere on the sphere, so $H_{ij}^\pm$ is positive. Otherwise, the following lemma would be useful:

Lemma. *Let $f(\theta,\phi)$ be a function defined on $\mathbb{S}^2$ that is odd in ϕ, and ρ be a positive function on $\mathbb{S}^2$ with*

$$\rho(\theta,\phi) = \rho(\theta, \phi + \tfrac{\pi}{2}) = \rho(\pi - \theta, \phi).$$

Assume that $f(\theta,\phi)$ is negative and strictly monotonically increasing in ϕ on the interval $(0,\pi)$. Then the integrals

$$K_1 = \int_{\mathbb{S}^2} f(\theta,\phi) f(\theta, \phi - \tfrac{\pi}{2}) \rho \, dA_{\mathbb{S}^2} < 0,$$
$$K_2 = \int_{\mathbb{S}^2} f(\theta,\phi) f(\pi - \theta, \phi - \tfrac{\pi}{2}) \rho \, dA_{\mathbb{S}^2} < 0.$$

Moreover, if $f(\theta,\phi) - f(\pi-\theta,\phi)$ is negative and monotonically increasing in ϕ on the interval $(0,\pi)$ for $0 < \theta < \frac{\pi}{2}$, then $K_1 < K_2 < 0$.

Proof. We decompose the sphere into four quadrants. Then

$$K_1 = \int_0^\pi \sin\theta \, d\theta \int_0^{\frac{\pi}{2}} \rho \, d\phi \big(f(\theta,\phi) f(\theta, \phi - \tfrac{\pi}{2}) + f(\theta, \phi + \tfrac{\pi}{2}) f(\theta,\phi)$$
$$+ f(\theta, \phi+\pi) f(\theta, \phi + \tfrac{\pi}{2}) + f(\theta, \phi - \tfrac{\pi}{2}) f(\theta, \phi + \pi) \big)$$
$$= \int_0^\pi \sin\theta \, d\theta \int_0^{\frac{\pi}{2}} \rho \, d\phi \big(f(\theta,\phi) + f(\theta, \phi + \pi) \big) \big(f(\theta, \phi - \tfrac{\pi}{2}) + f(\theta, \phi + \tfrac{\pi}{2}) \big)$$
$$= \int_0^\pi \sin\theta \, d\theta \int_0^{\frac{\pi}{2}} \rho \, d\phi \big(f(\theta,\phi) - f(\theta, \pi - \phi) \big) \big(- f(\theta, \tfrac{\pi}{2} - \phi) + f(\theta, \tfrac{\pi}{2} + \phi) \big)$$

For $0 < \phi < \frac{\pi}{2}$, by the assumed monotonicity, we have $f(\phi) < f(\pi - \phi) < 0$ and $g(\frac{\pi}{2} - \phi) < g(\frac{\pi}{2} + \phi) < 0$. So K_1 is negative. On the other hand,

$$\begin{aligned}
K_2 = \int_0^{\frac{\pi}{2}} \sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \rho\, d\phi \big(& f(\theta,\phi) f(\pi-\theta,\phi-\tfrac{\pi}{2}) + f(\pi-\theta,\phi) f(\theta,\phi-\tfrac{\pi}{2}) \\
& + f(\theta,\phi+\tfrac{\pi}{2}) f(\pi-\theta,\phi) + f(\pi-\theta,\phi+\tfrac{\pi}{2}) f(\theta,\phi) \\
& + f(\theta,\phi+\pi) f(\pi-\theta,\phi+\tfrac{\pi}{2}) + f(\pi-\theta,\phi+\pi) f(\theta,\phi+\tfrac{\pi}{2}) \\
& + f(\theta,\phi-\tfrac{\pi}{2}) f(\pi-\theta,\phi+\pi) + f(\pi-\theta,\phi-\tfrac{\pi}{2}) f(\theta,\phi+\pi) \big) \\
= \int_0^{\frac{\pi}{2}} \sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \rho\, d\phi & \\
\Big[& \big(f(\pi-\theta,\phi) + f(\pi-\theta,\phi+\pi)\big)\big(f(\theta,\phi-\tfrac{\pi}{2}) + f(\theta,\phi+\tfrac{\pi}{2})\big) \\
& + \big(f(\theta,\phi) + f(\theta,\phi+\pi)\big)\big(f(\pi-\theta,\phi-\tfrac{\pi}{2}) + f(\pi-\theta,\phi+\tfrac{\pi}{2})\big) \Big] \\
= \int_0^{\frac{\pi}{2}} \sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \rho\, d\phi & \\
\Big[& \big(f(\pi-\theta,\phi) - f(\pi-\theta,\pi-\phi)\big)\big(- f(\theta,\tfrac{\pi}{2}-\phi) + f(\theta,\phi+\tfrac{\pi}{2})\big) \\
& + \big(f(\theta,\phi) - f(\theta,\pi-\phi)\big)\big(- f(\pi-\theta,\tfrac{\pi}{2}-\phi) + f(\pi-\theta,\phi+\tfrac{\pi}{2})\big) \Big]
\end{aligned}$$

and the integrand is negative by the same argument as above.

Moreover,

$$\begin{aligned}
K_1 - K_2 = \int_0^{\frac{\pi}{2}} \sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \rho\, d\phi & \\
& \big(f(\theta,\phi) - f(\pi-\theta,\phi) - f(\theta,\pi-\phi) + f(\pi-\theta,\pi-\phi)\big) \\
& \big(- f(\theta,\tfrac{\pi}{2}-\phi) + f(\pi-\theta,\tfrac{\pi}{2}-\phi) + f(\theta,\tfrac{\pi}{2}+\phi) - f(\pi-\theta,\tfrac{\pi}{2}+\phi)\big).
\end{aligned}$$

Again the integrand is negative if $f(\theta,\phi) - f(\pi-\theta,\phi)$ is negative and monotonically increasing in ϕ on the interval $(0,\pi)$. So $K_1 < K_2$. □

Using this lemma, one easily proves that if p_i and p_j are horizontal neighbors (e.g. $(i,j) = (1,2)$) or on a vertical face diagonal (e.g. $(i,j) = (1,6)$), we have $H^{\pm}_{ij} < 0$. Consequently, by

$$\nabla_p^2 \log I^{\pm}(v,v) = -16\pi^2 \sum_{i\in A} \sum_{j \in B} H^{\pm}_{ij}$$

we can already conclude that the Hessian is positive definite on the irreps T_{1u}, E_g, but negative definite on E_u.

3.3. **The definiteness on T_{2u}.** For T_{2u} we place the cube so that a pair of antipodal vertices are at the north and the south pole. So

$$\begin{aligned}
p_1 &= (0,0), & p_5 &= (\pi,0), \\
p_2 &= (\theta_0,0), & p_6 &= (\pi-\theta_0,\pi/3), \\
p_3 &= (\theta_0,2\pi/3), & p_7 &= (\pi-\theta_0,\pi), \\
p_4 &= (\theta_0,4\pi/3), & p_8 &= (\pi-\theta_0,5\pi/3).
\end{aligned}$$

where $\theta_0 = \arccos(-1/3)$. Then a generator for T_{2u} is given by

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1,2,3,4\}, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{5,6,7,8\}.$$

Note that the vertices p_1 and p_5 (at the poles) are actually fixed under this generator. The Hessian becomes

$$\nabla_p^2 \log I^{\pm}(v,v) = -16\pi^2 \int_{\mathbb{S}^2} \partial_\phi G_A \partial_\phi G_B \,\mathrm{d}\mu^{\pm},$$

where $A = \{2,3,4\}$ and $B = \{6,7,8\}$. One easily notices that the integrand $\partial_\phi G_A \partial_\phi G_B \le 0$ everywhere on the sphere. So the Hessian is positive definite on T_{2u}.

3.4. **The definiteness on T_{2g}.** For T_{2g} we use a different orientation of the cube. Let $\Theta_0 = \arccos\sqrt{2/3}$, $a = \sqrt{2/3}$, $b = 1/\sqrt{3}$, and place the eight vertices so that

$$\begin{aligned} A &= \{(\pm b,\ 0,\ \pm a)\} \subset \mathbb{R}^3 \quad \text{(the } xz\text{-plane)},\\ B &= \{(\pm b,\ \pm a,\ 0)\} \subset \mathbb{R}^3 \quad \text{(the equatorial plane)}. \end{aligned}$$

In spherical coordinates these are

$$\begin{aligned} &p_1 = (\Theta_0, 0), && p_2 = (\Theta_0, \pi), && p_3 = (\pi - \Theta_0, 0), && p_4 = (\pi - \Theta_0, \pi),\\ &p_5 = \left(\tfrac{\pi}{2}, \tfrac{\pi}{2} - \Theta_0\right), && p_6 = \left(\tfrac{\pi}{2}, \tfrac{\pi}{2} + \Theta_0\right), && p_7 = \left(\tfrac{\pi}{2}, -\tfrac{\pi}{2} - \Theta_0\right), && p_8 = \left(\tfrac{\pi}{2}, -\tfrac{\pi}{2} + \Theta_0\right). \end{aligned}$$

A generator for T_{2g} is

$$v_i = \frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{1,2,3,4\} = A, \qquad v_i = -\frac{\partial_\phi}{\sin\theta}\Big|_{p_i} \text{ for } i \in \{5,6,7,8\} = B.$$

The Hessian becomes

$$\nabla_p^2 \log J_8(v,v) = -16\pi^2 \int_{\mathbb{S}^2} \partial_\phi G_A \,\partial_\phi G_B \,\mathrm{d}(\mu^+ + \mu^-).$$

Explicit formulas for $\partial_\phi G_A$ and $\partial_\phi G_B$. We work in Cartesian coordinates (X,Y,Z) on $\mathbb{S}^2$. Recall that $G_i(p) = -\frac{1}{2\pi}\log\sin\frac{\mathrm{dist}(p,p_i)}{2}$. For unit vectors p and p_i one has $\cos(\mathrm{dist}(p,p_i)) = p\cdot p_i$, so

$$\sin\frac{\mathrm{dist}(p,p_i)}{2} = \sqrt{\frac{1 - p\cdot p_i}{2}}, \qquad G_i = -\frac{1}{4\pi}\log(1 - p\cdot p_i) + C.$$

Derivation of G_A. The four A-vertices in Cartesian coordinates are $(\pm b, 0, \pm a)$, giving inner products $p\cdot p_i = \pm bX \pm aZ$ in all four sign combinations. Their sum yields

$$\begin{aligned} G_A = \sum_{i\in A} G_i = -\frac{1}{4\pi}\big[&\log(1 - bX - aZ) + \log(1 + bX - aZ)\\ &+ \log(1 - bX + aZ) + \log(1 + bX + aZ)\big] + C_A. \end{aligned}$$

Grouping in pairs,

$$(1 \mp bX - aZ)(1 \pm bX - aZ) = (1 - aZ)^2 - b^2X^2, \quad (1 \mp bX + aZ)(1 \pm bX + aZ) = (1 + aZ)^2 - b^2X^2,$$

so the product of all four factors is $P_A(X,Z) := \big((1 - aZ)^2 - b^2X^2\big)\big((1 + aZ)^2 - b^2X^2\big)$, and

$$G_A = -\frac{1}{4\pi}\log P_A(X,Z) + C_A.$$

In particular, G_A depends only on X and Z. An identical argument with $Z \to Y$ applies to the four B-vertices $(\pm b, \pm a, 0)$, giving $P_B(X,Y) := P_A(X,Y)$ and

$$G_B = -\frac{1}{4\pi}\log P_B(X,Y) + C_B.$$

Derivation of $\partial_\phi G_A$. The Killing field for rotation about the Z-axis is $\partial_\phi = X\partial_Y - Y\partial_X$ on $\mathbb{S}^2$. Since G_A is independent of Y, we have $\partial_Y G_A = 0$ and $\partial_\phi G_A =$

$-Y\,\partial_X G_A = \frac{Y}{4\pi}\frac{\partial_X P_A}{P_A}$. Writing $P_A = fg$ with $f = (1-aZ)^2 - b^2X^2$ and $g = (1+aZ)^2 - b^2X^2$, we get $\partial_X f = \partial_X g = -2b^2X$, so

$$\partial_X P_A = -2b^2X(f+g) = -2b^2X\big[2+2a^2Z^2-2b^2X^2\big] = -4b^2X(1+a^2Z^2-b^2X^2).$$

On $\mathbb{S}^2$ with $a^2 = 2/3$, $b^2 = 1/3$,

$$1 + a^2Z^2 - b^2X^2 = 1 + \tfrac{2}{3}Z^2 - \tfrac{1}{3}X^2 = \frac{3+2Z^2-X^2}{3},$$

so $\partial_X P_A = -\frac{4X(3+2Z^2-X^2)}{9}$, and

$$\partial_\phi G_A = \frac{Y}{4\pi}\cdot\frac{-4X(3+2Z^2-X^2)/9}{P_A} = -\frac{XY(3+2Z^2-X^2)}{9\pi P_A(X,Z)}.$$

Derivation of $\partial_\phi G_B$. Now G_B depends on both X and Y, so both terms of ∂_ϕ contribute:

$$\partial_\phi G_B = X\,\partial_Y G_B - Y\,\partial_X G_B = \frac{1}{4\pi P_B}\big(-X\,\partial_Y P_B + Y\,\partial_X P_B\big).$$

Writing $P_B = f_B g_B$ with $f_B = (1-aY)^2 - b^2X^2$ and $g_B = (1+aY)^2 - b^2X^2$, an identical argument gives

$$\partial_X P_B = -4b^2X(1+a^2Y^2-b^2X^2) = -\frac{4X(3+2Y^2-X^2)}{9}.$$

For $\partial_Y P_B$, we compute $\partial_Y f_B = -2a(1-aY)$ and $\partial_Y g_B = 2a(1+aY)$, so

$$\partial_Y P_B = 2a\big[(1+aY)f_B - (1-aY)g_B\big].$$

The bracket equals $(1+aY)(1-aY)\big[(1-aY)-(1+aY)\big] - b^2X^2\big[(1+aY)-(1-aY)\big] = (1-a^2Y^2)(-2aY) - 2ab^2X^2Y = -2aY(1-a^2Y^2+b^2X^2)$, and since $1-a^2Y^2+b^2X^2 = (3+X^2-2Y^2)/3$,

$$\partial_Y P_B = 2a\cdot(-2aY)\cdot\frac{3+X^2-2Y^2}{3} = -\frac{8Y(3+X^2-2Y^2)}{9}.$$

Substituting both into $\partial_\phi G_B$:

$$\begin{aligned}\partial_\phi G_B &= \frac{1}{4\pi P_B}\left[\frac{8XY(3+X^2-2Y^2)}{9} - \frac{4XY(3+2Y^2-X^2)}{9}\right]\\ &= \frac{XY}{36\pi P_B}\big[8(3+X^2-2Y^2) - 4(3+2Y^2-X^2)\big]\\ &= \frac{XY}{36\pi P_B}\cdot 12(1+X^2-2Y^2) = \frac{XY(1+X^2-2Y^2)}{3\pi P_B(X,Y)}.\end{aligned}$$

Hence the integrand equals

$$\partial_\phi G_A\cdot\partial_\phi G_B = \frac{N(X,Y,Z)}{27\pi^2 P_A(X,Z)\,P_B(X,Y)}, \qquad N = -X^2Y^2(3+2Z^2-X^2)(1+X^2-2Y^2).$$

Note that $3+2Z^2-X^2 = 2+Y^2+3Z^2 > 0$ on $\mathbb{S}^2$, and N vanishes at every branch point: at A-vertices $Y=0$, and at B-vertices $1+b^2-2a^2 = 1+\frac{1}{3}-\frac{4}{3} = 0$, so P_AP_B/N remains bounded there.

Symmetrization over the octahedral group. Let O_h be the 48-element symmetry group of the cube. Any $g \in O_h$ permutes the 8 vertices, so the product $P_A(X,Z)P_B(X,Y)$, which equals the product of the 8 squared half-chord-distance factors $(1 - p\cdot p_i)$, is O_h-invariant. The function $u = 2\pi\sum_i G_i$ is likewise O_h-invariant, so the weight $w\,\mathrm{d}A = \mathrm{d}(\mu^+ + \mu^-)$ is O_h-invariant. For each $g \in O_h$, a change of variables gives

$$\int_{\mathbb{S}^2}\frac{N(X,Y,Z)}{P_AP_B}\,w\,\mathrm{d}A = \int_{\mathbb{S}^2}\frac{N(g^{-1}(X,Y,Z))}{P_AP_B}\,w\,\mathrm{d}A.$$

Averaging over all $g \in O_h$ yields

$$\int_{\mathbb{S}^2} \frac{N}{P_A P_B}\, w\, \mathrm{d}A = \int_{\mathbb{S}^2} \frac{\bar{N}}{P_A P_B}\, w\, \mathrm{d}A, \qquad \bar{N} = \frac{1}{|O_h|} \sum_{g \in O_h} N \circ g^{-1}.$$

Since N is even in each coordinate, the 8 sign-change elements act trivially, and the average reduces to the average over the 6 permutations of (X, Y, Z).
Evaluating the symmetrized numerator. Set $p = X^2$, $q = Y^2$, $r = Z^2$ (so $p+q+r = 1$, $p, q, r \geq 0$). Each permuted term is $T(i, j, k) = -ij(3 + 2k - i)(1 + i - 2j)$. Expanding,

$$T(i, j, k) = -3ij - 2i^2 j + 6ij^2 - 2ijk - 2i^2 jk + 4ij^2 k + i^3 j - 2i^2 j^2.$$

Summing over the 6 permutations of (i, j, k) from (p, q, r) and applying the symmetric-polynomial identities

$$\sum ij = 2e_2, \quad \sum i^2 j = \sum ij^2 = e_2 - 3e_3, \qquad \sum ijk = 6e_3,$$
$$\sum i^2 jk = \sum ij^2 k = 2e_3, \qquad \sum i^3 j = e_2 - 2e_2^2 - e_3, \quad \sum i^2 j^2 = 2(e_2^2 - 2e_3),$$

where $e_2 = pq + qr + rp$ and $e_3 = pqr$, we obtain

$$\begin{aligned} 6\bar{N} = \sum_{\sigma \in S_3} T \ &= -3(2e_2) - 2(e_2 - 3e_3) + 6(e_2 - 3e_3) \\ &\quad - 2(6e_3) - 2(2e_3) + 4(2e_3) + (e_2 - 2e_2^2 - e_3) - 4(e_2^2 - 2e_3) \\ &= -(e_2 + 6e_2^2 + 13e_3). \end{aligned}$$

Hence

$$\bar{N} = -\frac{e_2 + 6e_2^2 + 13e_3}{6}.$$

Conclusion. Since $p, q, r \geq 0$, we have $e_2, e_3 \geq 0$, so $\bar{N} \leq 0$ everywhere on $\mathbb{S}^2$. Moreover $\bar{N} = 0$ only when $e_2 = 0$, i.e. at most one of X, Y, Z is nonzero, which on $\mathbb{S}^2$ occurs only at the six coordinate-axis poles $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, $(0, 0, \pm 1)$. These six points are not cube vertices, so $P_A P_B > 0$ and $w > 0$ there. Therefore $\bar{N}/(P_A P_B) \leq 0$ with equality only on a set of measure zero, and

$$\int_{\mathbb{S}^2} \partial_\phi G_A\, \partial_\phi G_B\, \mathrm{d}(\mu^+ + \mu^-) = \frac{1}{27\pi^2} \int_{\mathbb{S}^2} \frac{\bar{N}}{P_A P_B}\, w\, \mathrm{d}A < 0.$$

Hence $\nabla_p^2 \log J_8(v, v) > 0$, and the Hessian is positive definite on T_{2g}.

References

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